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Roll No.

220951

5th Sem/ Electrical
Subject : Electric Vehicle Technology

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Pure Electric vehicles is also known as (CO1)
- a) Battery electric vehicle
 - b) Hybrid Electric vehicles
 - c) Plug in Hybrid Electric vehicles
 - d) None of the above
- Q.2 FAME stands for (CO1)
- a) Faster addition and maintenance of Electric
 - b) Faster adoption and manufacturing of Electric
 - c) Feasibility addition & manufacturing of Electric
 - d) All of the above
- Q.3 BLDC motors are known for (CO2)
- a) High torque at low speed
 - b) Low torque at high speed
 - c) High efficiency at high speed
 - d) None of the above

- Q.4 What is the primary function of BMS. (CO3)
- a) To monitor and manage the battery pack
 - b) To adjust climate control
 - c) To control the vehicle speed
 - d) To control the regenerative braking
- Q.5 What is the main difference between a simple EV and a Hybrid EV. (CO4)
- a) Simple EVs use gasoline
 - b) Hybrid have no electric motor
 - c) Hybrid EVs can use both gasoline & Electricity
 - d) Simple EVs have no battery
- Q.6 DC fast charging is also known as which level of charging (CO3)
- a) Level 1
 - b) Level 2
 - c) Level 3
 - d) Level 4

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Writes the full form PHEVs. (CO1)
- Q.8 Writes two applications of solid state battery. (CO3)
- Q.9 Writes two advantages of PMSM motor. (CO2)
- Q.10 Define the ULTRA capacitor. (CO3)

- Q.11 Name two types of cooling system used to manage battery temp. in EVs? (CO4)
- Q.12 Define Hybrid Electric Vehicle. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Describe the importance of electric vehicles. (CO1)
- Q.14 Write short notes of FAME scheme? (CO1)
- Q.15 What are the advantages and disadvantages of BLDC motor. (CO3)
- Q.16 Discuss about the flywheel energy storage system in brief. (CO4)
- Q.17 What are the advantages and disadvantages of PMSM motor. (CO2)
- Q.18 Explain the safety precautions while handling a high voltage batteries in EV. (CO3)
- Q.19 What are the difference between battery electric vehicle and hybrid electric vehicles? (CO3)
- Q.20 Write the merit and demerits of Lithium-ion battery. (CO4)
- Q.21 Describe the control mechanism of BLDC motor. (CO2)
- Q.22 Discuss the components and working of an EV charger. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Discuss the working principle and control of permanent magnet synchronous motor (BLDC) used in electric vehicles. (CO2)
- Q.24 Explain the Govt. policies and regulation for encourage the adaptation and promotion of electric vehicles in the India. (CO1)
- Q.25 Describe the various charging and discharging test of Lithium-ion batteries. (CO4)

(**Note :** Course outcome/CO is for office use only)